Lesson notes · for you

Machines that notice

IoT · Level 1 · Lesson 1

Materials
Paper and a pencil for your table; Your eyes and a notebook on the way to school; An adult to ask when you're not sure what a device does

Learning goals

- Explain what a sensor does: it notices something and turns it into a number.
- Match your senses to sensors that do a similar job, and find senses machines have that you don't.
- Follow how a device senses, decides and acts.

Activity: Sensor hunt

1. Draw a table with four columns: the device, what it senses, what decides, and what it does. Leave room for at least eight rows.
2. At home, look for anything that switches, measures or warns by itself: a fridge, a solar lamp, a phone screen that dims, a smoke alarm.
3. At school and on the way, look at doors, taps, street lights, water tanks and buses. Write each one in your table.
4. Put a star by each sensor that notices something people can't: magnetism, very high sound, infrared, or a gas with no smell.
5. Choose your favourite find and explain its sense, decide and act steps to a friend or someone at home in under a minute.

Look, don't open: never take a device apart or touch sockets and mains wires. Ask before you photograph something that isn't yours.

Discussion

- Which of your senses would be hardest for a machine to copy, and why?
- When should a person, not a sensor, make the decision?
- What would you want a sensor in your town to warn people about?

Check yourself

- Name one device near you, and say what it senses, what decides and what it does.

Try it this week
Go on a sensor hunt: find eight things that switch by themselves, and write what each senses, what decides and what it does.

Worksheet

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Name _____ Class _____ Date _____

1 **Your school's water tank has a sensor that notices how full it is. When it reads low, the pump starts. Which part made the decision?**

- ◯ The pump, because it's the part that moves
- ◯ The sensor, because it noticed the water was low
- ◯ A small computer following the rule it was given

2 **Match each job to the sensor that does it.**

Turning a map to face north	• • • •	A carbon monoxide alarm
Warning a driver that a wall is close behind	• • • •	An infrared thermometer
Taking a temperature without touching	• • • •	A light sensor
Warning of a gas with no smell	• • • •	A compass sensor (magnetism)
Switching a street light on at dusk	• • • •	A parking sensor (sound too high to hear)

3 **A loudspeaker is a sensor, because it works with sound.**

◯ True ◯ False

4 **An automatic street light works through the night. Put its steps in order.**

- ☐ The controller compares it with its rule: dark enough?
- ☐ Daylight fades in the evening
- ☐ The lamp switches on
- ☐ At dawn the reading rises, and the lamp switches off
- ☐ The light sensor's reading drops

5 **Which of these sense something by themselves?**

- ☐ A loudspeaker
- ☐ A light switch on the wall
- ☐ A smoke alarm
- ☐ A fridge that keeps itself cool
- ☐ A torch you switch on
- ☐ A hand dryer that starts when your hands are under it

✎ **Which sensor around you do you rely on without noticing? What would happen if it stopped working?**

Check your answers

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1 **A small computer following the rule it was given**

The sensor only measures and the pump only acts.

2 **Turning a map to face north — A compass sensor (magnetism)**
Warning a driver that a wall is close behind — A parking sensor (sound too high to hear)
Taking a temperature without touching — An infrared thermometer
Warning of a gas with no smell — A carbon monoxide alarm
Switching a street light on at dusk — A light sensor

Each sensor notices one thing. The first four notice things people can't: magnetism, very high sound, an exact temperature from a distance and a gas with no smell.

3 **False**

A loudspeaker makes sound: it acts, so it's an actuator.

4 **Top to bottom: 3, 1, 4, 5, 2**

Sense, decide, act, again and again: the reading changes first, the controller decides, and the lamp acts.

5 **A hand dryer that starts when your hands are under it; A smoke alarm; A fridge that keeps itself cool**

The dryer senses your hands, the smoke alarm senses smoke, and the fridge senses its own temperature.